# Alyssa Dang

adang2@andrew.cmu.edu | linkedin.com/in/alyssakdang | github.com/alyssahui | alyssahui.github.io

## EDUCATION

**Carnegie Mellon University, Pittsburgh, PA** *Expected May 2028*

*B.S. in Human-Computer Interaction + Cognitive Science (Computer Science) | GPA: 4.0/4.0*

- Relevant Coursework: Computational Neuroscience, Designing Human-Centered Software (in progress), Principles of Imperative Computation (in progress), Cognitive Psychology
- Activities: Scottylabs Design, CMU Mock Trial, Game Creation Society

## RESEARCH & TECHNICAL EXPERIENCE

**Research Assistant, Laboratory of Computational Perception** *Aug 2026 - Present*

- Collaborate with a small research team under faculty supervision to train deep neural networks in PyTorch, studying how task objectives shape the models' learned representations of body position and movement

**V1 Neural Decoding & Systems Identification, Computational Neuroscience Lab** *Apr 2026*

- Analyzed mouse primary visual cortex (V1) Neuropixel recordings (Allen Institute/DANDI) across drifting-grating and Gabor stimuli using Gaussian-kernel smoothing, tuning curves, spatial receptive field mapping, and Linear-Nonlinear-Poisson (LNP) model validation via spike-triggered averaging
- Applied Principal Component Analysis (PCA) via SVD on zero-centered population response matrices, identifying a low-dimensional manifold where 6 principal components captured 95% of variance
- Developed OLS linear regression decoders with circular target encodings (80/20 train/test split), demonstrating significantly higher decoding accuracy for stimulus orientation over direction (31° vs. 67° mean test error)

**LTP/LTD Circuit Simulation, Computational Neuroscience Lab** *Feb 2026*

- Built a Leaky Integrate-and-Fire (LIF) circuit model in Python (NumPy, Matplotlib, Jupyter Notebook) simulating AMPA/NMDA receptor kinetics, calcium current, and membrane voltage dynamics of an excitatory synapse
- Implemented Spike-Timing-Dependent Plasticity (STDP) via calcium-threshold learning rules modulating synaptic conductance (LTP/LTD) within a biologically accurate 20ms spike-timing window
- Validated model behavior against published literature through comparative plots of I-V curves and conductance history

## PROJECTS

**NBA Outcome Prediction & Optimization Model, ML & Probabilistic Modeling** *Apr 2026*

- Modeled NBA home win probabilities in R using Random Forest classifiers across multiple architectures (all predictor variables, top predictors, and home team predictors)
- Evaluated models via 5-fold cross-validation, achieving a 0.102 Brier score on top feature sets, and quantified prediction uncertainty via tree standard deviations to guide expected-value decision-making

## ADDITIONAL EXPERIENCE

**Product & UI/UX Designer, CravingChef (iOS App Store)** *Jul 2026 – Aug 2026*

- Partnered directly with the founder to shape product decisions as UI/UX designer, synthesizing user feedback and behavioral analytics into interface decisions and establishing human-centered research workflows applicable to neurotechnology and assistive human-machine interfaces

**Project Lead, Honey Haven — CMU Game Creation Society** *Jan 2026 – Apr 2026*

- Directed a small cross-functional team (design, narrative, software) to pitch, scope, and ship a project in one semester; won Game Creation Society Gold Award

## SKILLS

**Technical:** Python, PyTorch, NumPy, Matplotlib, R, C, Dimensionality Reduction (PCA, SVD), Linear Regression Decoders, LNP Modeling, LIF Neurons, Spike-Triggered Averaging, Jupyter, HTML/CSS, JavaScript, Git Command Line

**Research & Signal Processing:** Neural Data Processing (Neuropixels/DANDI), Signal Filtering, Receptive Field Mapping, Computational Modeling, Cross-Validation, Usability Testing

**Collaboration & Product:** Figma, Cross-Functional Collaboration, Stakeholder Communication, Google Workspace, Adobe Creative Suite